

Exact Radial-Determinant Frequencies of Annular and Circular Plates: the Closed-Ring and Solid-Disk Limits of the Sector Wave-Function Method

Garrek McCune^{a,*}, Daniel Stutts^a

^a*Mechanical and Aerospace Engineering,*

Missouri University of Science and Technology, Rolla, MO 65409, USA

*Corresponding author. E-mail: ghmkfh@umsystem.edu

DRAFT — September 9, 2026

Abstract

The Seok–Tiersten–Scarton wave-function method for annular-sector plates has two classical limits that never form the two-edge Galerkin determinant. At sector angle $2\Theta = 2\pi$ there are no radial walls: circumferential order is an integer n , and each frequency is a zero of a 4×4 radial edge-determinant already present in the sector assembler. At inner radius $R_i = 0$ the two origin-singular branches drop out and the same operator becomes a 2×2 at the outer radius — the classical Kirchhoff disk. Because there are no θ -edges, the Kirchhoff corner jump that blocks simply supported sectors never appears, and the nine inner/outer edge combinations of Vogel and Skinner become different row pairs of one matrix. Frequencies are found from $\det L(n, \Omega) = 0$ from scratch; published tables enter only post hoc. A frozen sign-flip ladder (inner five of five real-part flips, with positive and negative controls in the same run) replaces the sector residual bar, which does not travel to this geometry. Out-of-plane free-free annuli recover NASA SP-160 Table 2.35 at 80/80 (MATCH + WEAK) across five radius ratios and both $\nu = 0.30$ and $\nu = 1/3$, and Narita’s four five-figure hold-outs at $b/a = 0.3$. The solid-disk 2×2 recovers Narita Table 2(b) at 4/4 and SP-160 Table 2.5 at 6/6; a naive 4×4 at $R_i = 0$ has rank three for $n \geq 1$ and is not the disk. Mixed-edge Tables 2.22 and 2.30 pass at 80/80 and 72/76, the $n = 0$ column at $b/a = 0.9$ of Table 2.30 being a named source defect internally inconsistent with Vogel’s own $n = 1$ column. Southwell’s Table 2.31 scores 13/18 under the same bar; the five direct misses are a sensitivity to the printed radius ratio rather than a solver defect, and independent SHELL281 finite elements at $b/a = 0.840$, $n = 0$ give $\lambda^2 = 132.549$ against the solver’s 132.990 (+0.333%) and Southwell’s printed 81.0 (−63.6%). Eighteen solver-versus-FE points across three mixed-edge geometries agree to better than 0.8% with a one-sided Kirchhoff-versus-Mindlin offset that grows monotonically in n . In-plane free-free rings recover Irie, Yamada and Muramoto’s Table 2 at 48/48. Default clamped-free sector assembly is unchanged.

Keywords: annular plate; circular plate; free vibration; exact solution; radial determinant; mixed boundary conditions; in-plane vibration.

1 Introduction

Annular and circular plates are the $2\Theta = 2\pi$ and $R_i = 0$ limits of the annular-sector geometry treated in the companion paper [11]. Classical Kirchhoff tables for those limits have existed for

decades: Leissa’s monograph [5] compiles the exact annular results of Vogel and Skinner [6], Southwell’s inner-clamped annulus as compiled in SP-160 Table 2.31 [7], and the axisymmetric work of Raju [8]; Narita [9] added five-figure free-free values for both the annulus and the solid disk; Irie, Yamada and Muramoto [10] tabulated the in-plane annular spectrum. The contribution here is not another Bessel catalogue. It is the reduction of a *validated sector solver* onto those tables, in both motions, with a screen that does not use the sector residual bar, together with the mixed inner/outer edge combinations that the sector corner problem blocks.

Seok, Tiersten and Scarton [1, 2, 3, 4] gave exact-wave-function solutions for annular-sector and rectangular plates, out-of-plane and in-plane, at clamped-free. The first companion paper [11] extended the annular-sector flexural assembler to completely free sectors and built a residual screen against independent finite elements; the second [12] did the rectangular analogue, where the Kirchhoff corner jump is load-bearing. Neither takes the 2π or $R_i = 0$ limit as a production entry point. That limit is cheaper than the sector: circumferential order is an integer, the characteristic matrix is 4×4 (ring) or 2×2 (disk), there are no θ -walls, and there is no corner jump. It is also the elastic forward model that a later piezoelectric identification will need on the free-free and clamped-free ring.

The detector remains paper-faithful. Frequencies are zeros of $\det L(n, \Omega) = 0$ built from the same Frobenius radial content as the sector assembler. Published values never seed a search; they enter only when a converged root is scored against them. The sign-flip ladder described in §3 was frozen before the production sweeps and was not retuned to admit a published match. Independent cyclic-symmetry SHELL281 calculations [13] are a third method, not a fitted target.

What this paper is not: McGee’s solid *sector* (that is the sector assembler with a coordinate singularity); elliptical plates; simply supported or guided sector edges (the open corner-jump problem); piezoelectric coupling, which appears only as one sentence of future work; or a splice into either companion paper.

2 Reductions

2.1 Closed ring, out-of-plane

Requiring 2π -periodicity of the transverse displacement forces the circumferential order to an integer, $n \in \mathbb{N}_0$. For $n \geq 1$ the $\cos n\theta$ and $\sin n\theta$ solutions are degenerate and share a frequency; for $n = 0$ the mode is a singleton. Rigid-body motions on a completely free ring are $n = 0, s = 0$ (translation) and $n = 1, s = 0$ (rotation). They are excluded from the elastic tables here, as in both companion papers.

The radial operator is the 4×4 characteristic matrix already assembled for the sector at a fixed separation constant. Its rows are, in order,

$$0 = M_r(\text{inner}), \quad 1 = M_r(\text{outer}), \quad 2 = V_r(\text{inner}), \quad 3 = V_r(\text{outer}), \quad (1)$$

and searching $\det L(n, \Omega) = 0$ at fixed integer n is the closed ring: no θ -edge condition is weakly enforced, so no residual is generated and no residual bar is meaningful. The four Frobenius branches span $\{J_n, Y_n, I_n, K_n\}$ under the radial map used throughout the series (Supplementary Material, §S.1).

Boundary conditions enter as row swaps on (1):

- **Free (F).** Leave that edge’s M_r and V_r rows in place.
- **Clamped (C).** Replace *both* of that edge’s rows by W and W' evaluated at the same radius.

- **Simply supported (S).** Replace only the V_r row by W , leaving M_r . This case is admitted by the same construction but is not exercised in the production tables below.

Two conventions must be kept straight. Leissa’s table titles list the *outer* edge first, while the interface used here is $(bc_{\text{inner}}, bc_{\text{outer}})$; SP-160 Table 2.22 (“Clamped, Free”) is therefore inner F / outer C, and Tables 2.30 and 2.31 (“Free, Clamped”) are inner C / outer F. Clamped–clamped is admitted by the same row map and was checked against the clamped–clamped strip limit $\lambda^2 \rightarrow 22.3733 (a/w)^2$ for a thin ring of width w , giving ratios 0.9999, 1.0000 and 1.0000 at $b/a = 0.90$, 0.94 and 0.97; it is not among this paper’s production tables.

2.2 Solid disk, out-of-plane

Boundedness at $r = 0$ removes Y_n and K_n . Two surviving coefficients and two outer-edge conditions give a 2×2 characteristic matrix. The practical difficulty is that the four series columns are mid-radius Taylor initial conditions rather than the Bessel pair itself, and a truncated series is finite at the origin, so “smallest $|W|$ at $r = 0$ ” does not isolate the regular branches. Regular solutions behave as $r^{|n|}$, which supplies two homogeneous conditions — an inner jet — whose two-dimensional kernel *is* the regular pair:

$$n = 0 : W' = W''' = 0; \quad n = 1 : W = W'' = 0; \quad n \geq 2 : W = W' = 0. \quad (2)$$

Projecting the two outer rows of the 4×4 onto that kernel gives the production 2×2 (§S.2).

A documented failure is worth naming, because it is the natural thing to try. Evaluating the 4×4 at $R_i = 0$ and looking for its zeros gives a matrix of rank 2 at $n = 0$ but rank 3 — not 2 — for $n \geq 1$ (Table 6). Frequencies extracted from that matrix are not Leissa’s disk. The production path is the 2×2 . A tiny-hole 4×4 at $b/a = 10^{-3}$ is a check on that path, not the path itself, and it is a check the implementation fails on two of four rows against a pre-registered 10^{-3} relative bar (§5). That bar is not loosened.

2.3 In-plane ring

The in-plane characteristic matrix is N_r and $N_{r\theta}$ at both arcs, and the same substitution applies: fix $\xi = n$ at an integer, search Ω . There is no Kirchhoff corner term to reduce. The in-plane frequency parameter of Irie, Yamada and Muramoto is defined from the extensional rigidity $D_{\text{ext}} = Eh/(1-\nu^2)$ as $\lambda^2 = \rho h a^2 \omega^2 / D_{\text{ext}}$, that is

$$\lambda = \omega a \sqrt{\rho(1-\nu^2)/E}, \quad (3)$$

which is a different symbol from the flexural λ^2 of (4) and must not reuse the flexural conversion factor. Their “circular” column is $\beta = b/a = 0.01$, a tiny-hole control, and is not claimed here as $R_i = 0$.

2.4 Conversion

All flexural comparisons use

$$\lambda^2 = \omega a^2 \sqrt{\rho/D}, \quad a = R_o, \quad D = \frac{Eh^3}{12(1-\nu^2)}, \quad (4)$$

with h the full thickness. The map from the solver’s native Ω to (4) is recomputed from the geometry and material of each run and is *not* a universal constant. On the free–free steel plate used as the fidelity anchor in [11] ($R_o = 8$ m, thickness 0.08 m, $b/a = 0.5$) the factor happens to be

$4\pi^2 \approx 39.478$; at $b/a = 0.3$ it is a different number, and reusing it is an error the series has made once and now regression-tests against. The unit test is the anchor’s mode 7: native $\Omega = 1.369611$ converts to $\lambda^2 = 54.0755$, matching the finite-element value to 0.01%. Native Ω is never printed with a λ^2 label.

Poisson ratios are not mixed in a single comparison. Table 2.34 states $\nu = 1/3$ in its caption and is compared only at that value. Narita and the finite-element models use the ν printed in their own captions (0.30 or 0.33); Irie’s table is $\nu = 0.3$. Tables 2.35, 2.22 and 2.30 print no Poisson ratio, so every sweep that touches those tables was run at both $\nu = 0.30$ and $\nu = 1/3$. The paper-facing comparison for Table 2.35 is the $\nu = 0.30$ column; Table 2.35’s ν is not asserted to be the $1/3$ of Table 2.34.

3 Screen

The depth-and-gap residual bar used for completely free *sectors* [11] does not travel to this geometry: applied unchanged it rejects five genuine full-ring roots. The ring and disk therefore use a different instrument, fixed before the production sweeps were launched:

1. A coarse scan of $\log_{10} |\det L|$ on a window local to each literature target, *and* a gap-free coarse scan per n . Both, not either: the free-free wells at the anchor geometry are narrower than 5×10^{-4} in Ω , and a target-local window alone once missed a whole band.
2. Golden-section polish of each candidate minimum.
3. A real-part sign-flip ladder at $\delta = 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}$ about the polished root. CLEAN means all five inner flips are present. FLOOR means the only inner miss is at $\delta = 10^{-6}$ with all five outer flips present and the residual shrinking — the arbitrary-precision working floor at 40 digits, already named in [11]. NOT means any inner miss at $\delta \geq 10^{-5}$, or a broken outer/shrinking pattern.
4. Positive controls must come back CLEAN and negative controls must come back NOT, evaluated in the same run as the production rows.

Scoring against a published number is separate from the ladder and is applied to the relative miss alone: MATCH below 1%, WEAK below 3%, MISS otherwise. FLOOR never converts a MATCH into a MISS. The ladder is never retuned to include a published value, and the 1%/3% thresholds are not adjusted table by table. This screen is neither the persistence screen of [12] nor the sector subdomain instrument of [11]; those names do not apply here.

A search window is a hypothesis. If $\log |\det L|$ falls monotonically to an endpoint of the scanned window, the minimiser can return that endpoint. The returned value is then a property of the window, not of the plate: it is n -independent and meaningless. The sign-flip ladder catches this — every such case in this work came back NOT — but the MATCH/WEAK/MISS gate above, computed from the relative miss alone, does not. A post-hoc label (EXACT if the returned Ω is a window endpoint, NEAR if within one coarse step of one, NONE otherwise) is therefore attached to every result. The label only labels: it never moves the returned Ω , so no computed frequency changes and the solver’s checkpoint version key is unaffected. An audit of all 342 stored rows of this study found 12 endpoint returns and no near-misses, every one of them in the mixed-edge sweep of §6, which is the only sweep combining a steep $d\lambda^2/d(b/a)$ with a fixed absolute half-window. Those cells are re-searched on a wide window where a root exists, and are marked in the tables below. Endpoint values are not frequencies and are not printed as such anywhere in this paper.

4 Out-of-plane free-free annulus

4.1 SP-160 Table 2.35

Table 2.35 of [5], from Vogel and Skinner [6], gives eight (n, s) rows at five radius ratios. Every one of the 80 cells across both Poisson ratios is MATCH or WEAK. At $\nu = 0.30$ (Table 1) all 40 cells are MATCH, the largest miss being 0.454% at $(1, 1)$, $b/a = 0.1$, against a three-figure printed value. At $\nu = 1/3$ (Supplementary Material, §S.4) 20 cells are MATCH and 20 are WEAK, the largest miss 2.220% at $(2, 1)$, $b/a = 0.9$. The WEAK cells at $\nu = 1/3$ cluster on the $(2, 0)$ and $(3, 0)$ branches, and those printed values coincide with SP-160 Table 2.34, which is Raju’s $\nu = 1/3$ set [8], not Vogel’s Table 2.35. Since Table 2.35 prints no Poisson ratio, the paper-facing comparison for this table is the $\nu = 0.30$ column, and Table 2.35’s ν is not asserted to be the $1/3$ of Table 2.34.

Table 1: Free-free annulus, $\nu = 0.30$. Printed values are SP-160 Table 2.35 [5, 6]; solver values are λ^2 from $\det L(n, \Omega) = 0$ on the 4×4 of §2.1. All 40 cells MATCH ($< 1\%$); largest miss 0.454%. Rigid-body $n = 0, 1$ at $s = 0$ are excluded.

(n, s)	$b/a = 0.1$		$b/a = 0.3$		$b/a = 0.5$		$b/a = 0.7$		$b/a = 0.9$	
	printed	solver	printed	solver	printed	solver	printed	solver	printed	solver
(2, 0)	5.30	5.3039	4.91	4.9060	4.28	4.2711	3.57	3.5725	2.94	2.9351
(3, 0)	12.4	12.437	12.26	12.266	11.4	11.425	9.86	9.8589	8.14	8.1449
(0, 1)	8.77	8.7747	8.36	8.3534	9.32	9.3135	13.2	13.163	34.9	34.834
(1, 1)	20.5	20.407	18.3	18.292	17.2	17.198	22.0	21.914	55.7	55.719
(2, 1)	34.9	34.930	33.0	32.973	31.1	31.115	37.8	37.841	93.8	93.794
(3, 1)	53.0	52.965	51.0	51.005	47.4	47.462	55.7	55.701	135.0	135.41
(0, 2)	38.2	38.236	50.4	50.353	92.3	92.308	251.0	250.58	2238.0	2238.9
(1, 2)	59.0	59.072	58.8	58.784	96.3	96.266	253.0	253.13	2240.0	2240.8

4.2 Narita’s five-figure hold-out

Table 2.35 is a three-figure table, so it cannot discriminate at the level the determinant actually resolves. Narita’s Table 2(a) [9] at $b/a = 0.3$, $\nu = 0.3$ is the hold-out: four five-figure values that were not used in any development decision. All four are recovered to between 3.7×10^{-6} and 1.6×10^{-5} relative (Table 2); the printed five figures hold in every case. The $(3, 0)$ row is FLOOR rather than CLEAN — its only inner miss is at $\delta = 10^{-6}$, with all five outer flips present and shrinking — which is the precision floor of the ladder, not a missing frequency.

Table 2: Narita Table 2(a) [9] five-figure hold-out, free-free annulus, $b/a = 0.3$, $\nu = 0.3$.

(n, s)	printed λ^2	solver λ^2	rel. miss	ladder
(2, 0)	4.9060	4.9060	6.4e-06	CLEAN
(0, 1)	8.3535	8.3535	3.7e-06	CLEAN
(3, 0)	12.266	12.266	1.6e-05	FLOOR
(1, 1)	18.292	18.292	1.1e-05	CLEAN

4.3 Fidelity anchors and finite elements

Five free-free roots at $b/a = 0.5$, $\nu = 0.30$ are carried forward from [11] as a fidelity gate on the production interface rather than as a new result: the ring path must reproduce them in native Ω before any table is believed. It does, to between 2.9×10^{-6} and 1.4×10^{-5} (Table 3).

Independent cyclic-symmetry SHELL281 models at the same geometry give the elastic listing 4.2645, 9.3114, 11.4082, 17.0973, 21.0296, agreeing with the solver to between 0.023% and 0.591%. Mesh convergence at that geometry is 4.264506, 4.264461, 4.264453 for 64, 96 and 128 circumferential divisions — a relative change of 10^{-5} between the two finest meshes. Contour plots of U_Z confirm the published (n, s) assignment on every mode in the plotted set, including the (3, 0), (1, 1) and (4, 0) rows whose labels had previously been inferred from frequency ordering alone. At $b/a = 0.9$ the finite-element modes 15–16 are in-plane ($|U_Z| \sim 10^{-13}$) and must not be given a flexural n ; the flexural (0, 1) mode at that geometry is the first axisymmetric elastic mode above that pair.

Table 3: Free-free annulus at $b/a = 0.5$, $\nu = 0.30$: the five fidelity anchors of [11] in native Ω , and the same modes against independent SHELL281 [13] in λ^2 . Ladder FLOOR here is the 40-digit working floor, not a missing root.

(n, s)	anchor Ω	solver Ω	rel.	ladder	FE λ^2	solver λ^2	rel. (%)
(2, 0)	0.1081885366	0.1081900894	1.4e-05	FLOOR	4.2645	4.2711	0.155
(0, 1)	0.2359132157	0.2359147685	6.6e-06	FLOOR	9.3114	9.3135	0.023
(3, 0)	0.2894098129	0.2894082601	5.4e-06	FLOOR	11.4082	11.4254	0.151
(1, 1)	0.4356362253	0.4356346725	3.6e-06	FLOOR	17.0973	17.1983	0.591
(4, 0)	0.5336384799	0.5336400327	2.9e-06	FLOOR	21.0296	21.067	0.178

5 Out-of-plane free disk

The solid disk uses the 2×2 of §2.2. Narita’s Table 2(b) [9] at $\nu = 0.33$ is again the five-figure hold-out and is recovered 4/4 at between 1.7×10^{-6} and 9.4×10^{-6} relative (Table 4). The (2, 0) hold-out is MATCH on the relative miss and NOT on the sign-flip ladder; it is trusted because the finite elements sit on the same value (5.2584 against the solver’s 5.2620, 0.069%). SP-160 Table 2.5 [5], six rows at the same Poisson ratio, is recovered 6/6 (Table 5); its largest disagreement is the (2, 0) row, printed 5.253 against the solver’s 5.262, a 0.172% difference that sits inside the 1% bar appropriate to a four-figure table and is resolved in the solver’s favour by the five-figure hold-out and by the finite elements, both of which give 5.262 and 5.2584 respectively. The Table 2.5 (0, 1) row is likewise MATCH on error and NOT on the ladder, and is read the same way.

Independent SHELL281 models of the free disk at $\nu = 0.33$ give 5.2584, 9.0679, 12.2284 and 20.5010 for the four Narita modes: 0.068%, 0.011%, 0.13% and 0.058% from the printed values and 0.069%, 0.011%, 0.13% and 0.059% from the 2×2 . Contours confirm the published (n, s) on all six Table 2.5 rows.

Table 4: Free solid disk, $\nu = 0.33$, from the 2×2 of §2.2, against Narita Table 2(b) [9] (five-figure hold-out) and independent SHELL281 [13].

(n, s)	printed λ^2	solver λ^2	rel. miss	FE λ^2	solver vs FE (%)	ladder
(2, 0)	5.262	5.2620	2.9e-06	5.2584	0.069	NOT
(0, 1)	9.0689	9.0689	1.7e-06	9.0679	0.011	FLOOR
(3, 0)	12.244	12.244	9.4e-06	12.2284	0.127	CLEAN
(1, 1)	20.513	20.513	4.7e-06	20.5010	0.059	CLEAN

Table 5: Free solid disk, $\nu = 0.33$, against SP-160 Table 2.5 [5]. All six MATCH. The $(2, 0)$ row is the largest disagreement at 0.172%; the five-figure hold-out and the finite elements both sit on the solver’s value.

(n, s)	printed λ^2	solver λ^2	rel. (%)	FE λ^2	ladder
$(2, 0)$	5.253	5.2620	0.172	5.2584	CLEAN
$(3, 0)$	12.23	12.244	0.114	12.2284	CLEAN
$(4, 0)$	21.6	21.527	0.337	21.4890	CLEAN
$(0, 1)$	9.084	9.0689	0.166	9.0679	NOT
$(1, 1)$	20.52	20.513	0.034	20.5010	CLEAN
$(2, 1)$	35.25	35.243	0.021	35.1988	CLEAN

5.1 Why the 4×4 at $R_i = 0$ is not the disk

Substituting $R_i = 0$ into the ring 4×4 and looking for its zeros is the obvious shortcut and it is wrong. Table 6 lists the ordered $\log_{10}$ singular values of that matrix at a representative frequency: at $n = 0$ there are two small values and the rank is 2, but at $n = 1, 2, 3$ only one singular value collapses and the rank is 3. A rank-3 4×4 has a one-dimensional kernel that is not the two-dimensional regular subspace, and its determinant zeros are not the plate’s frequencies. This is the reason the production disk path is the 2×2 and why that fact is enforced by a regression test rather than by convention.

Table 6: Ordered $\log_{10}$ singular values of the ring 4×4 evaluated at $R_i = 0$, with the observed and expected numerical rank. Rank 3 for $n \geq 1$ is why this matrix is not the disk.

	σ_1	σ_2	σ_3	σ_4	rank	expected
$n = 0$	0.43	-0.96	-16.09	-31.88	2	2
$n = 1$	0.60	-2.41	-2.88	-15.51	3	3
$n = 2$	0.60	-1.38	-3.81	-16.94	3	3
$n = 3$	0.60	-1.89	-4.91	-16.61	3	3

5.2 A named failure: the tiny-hole control

A separate control asks whether the 4×4 at a very small hole, $b/a = 10^{-3}$, reproduces the 2×2 disk to a pre-registered 10^{-3} relative tolerance. It does not (Table 7): $(2, \cdot)$ and $(0, \cdot)$ differ by 0.940% and 0.545%, and the control is reported as a FAIL. The hole roots are real and deep, and both failing rows have ladder NOT while the two passing rows are CLEAN, so this is the residual physical difference between a punctured and a solid plate at that hole size, not a missed disk frequency. The tolerance is not loosened to make the control pass.

Table 7: Tiny-hole control, $b/a = 10^{-3}$, $\nu = 0.33$: the 4×4 at a small hole against the production 2×2 . Pre-registered bar 10^{-3} relative; the control is a named FAIL and the bar is not loosened.

mode	disk 2×2	hole 4×4	rel. (%)	ladder	control
$(2, \cdot)$	5.2620	5.3115	0.940	NOT	FAIL
$(0, \cdot)$	9.0689	9.1183	0.545	NOT	FAIL
$(3, \cdot)$	12.244	12.245	0.007	CLEAN	PASS
$(1, \cdot)$	20.513	20.524	0.054	CLEAN	PASS

6 Mixed inner and outer edges

Mixed inner/outer conditions are the content of this paper that is not a free-free echo of the companion papers, and they are reached by the row swap of §2.1 with no new derivation. Clamping one arc of a ring is also the mechanical boundary condition of a stator disk; that observation is a statement about the boundary condition only, and no electromechanical claim is made here.

6.1 Vogel and Skinner, Tables 2.22 and 2.30

Table 2.22 (inner F, outer C) is recovered at 80/80 across both Poisson ratios and all five radius ratios including $b/a = 0.9$, with 78 cells MATCH and 2 WEAK (Table 8; the $\nu = 1/3$ half is §S.4). The thin-ring column is worth naming explicitly: the 4×4 does not degrade at $b/a = 0.9$, where the printed values run to 6183 and the solver returns 6185.1.

Table 8: Mixed edges, inner free / outer clamped, $\nu = 0.30$. Printed values are SP-160 Table 2.22 [5, 6] (Leissa’s title lists the outer edge first). 80/80 MATCH + WEAK over both Poisson ratios.

(n, s)	$b/a = 0.1$		$b/a = 0.3$		$b/a = 0.5$		$b/a = 0.7$		$b/a = 0.9$	
	printed	solver	printed	solver	printed	solver	printed	solver	printed	solver
(0, 0)	10.2	10.159	11.4	11.424	17.7	17.715	43.1	43.142	360	360.35
(1, 0)	21.1	21.195	19.5	19.540	22.0	22.015	45.3	45.333	362	361.50
(2, 0)	34.5	34.535	32.5	32.594	32.0	32.115	51.5	51.585	365	364.96
(3, 0)	51.0	50.989	49.1	49.069	45.8	45.812	61.3	61.289	370	370.70
(0, 1)	39.5	39.521	51.7	51.745	93.8	93.847	253	251.64	2219	2219.5
(1, 1)	60.0	60.061	59.8	59.759	97.3	97.376	254	253.72	2220	2220.9
(2, 1)	83.4	83.478	79.0	79.061	108.0	107.49	259	259.91	2225	2225.2
(0, 2)	90.4	90.445	132.0	132.41	253.0	252.19	692	692.01	6183	6185.1

Table 2.30 (inner C, outer F) is reported here as 72/76 (Table 9). Four cells are excluded as one named source defect and four are direct misses. Both need justifying.

The excluded cells are the whole $n = 0$ column at $b/a = 0.9$, at both Poisson ratios. At that radius ratio the thin-ring $n = 0$ and $n = 1$ branches are nearly degenerate, and Vogel’s *own* Table 2.22 at the same b/a prints that degeneracy: 360 against 362 for $s = 0$, and 2219 against 2220 for $s = 1$. Table 2.30 at the same b/a instead prints 51.5 against 345 and 970 against 2189. The $n = 1$ entries agree with the 4×4 to 0.06%; the $n = 0$ entries agree with nothing, and no confident root exists anywhere in $\lambda^2 \in [15, 155]$ at the first of them. The argument for excluding that column is therefore internal to the source — Vogel’s other table plus the physical degeneracy — and not “the solver disagrees”. The MATCH/WEAK bar is unchanged. For the record, the production sweep scored this table 72/78, excluding only the printed 51.5; the paper’s denominator differs from the sweep’s by treating the column as one skip rather than one cell.

The four direct misses are the (1, 0) entries at $b/a = 0.1$ and $b/a = 0.3$, at both Poisson ratios: printed 3.14 against solver 3.478 (10.8%) and printed 6.33 against solver 6.552 (3.5%). They are named, not dropped, and they remain open: both lie on the $n = 1$, $s = 0$ branch at the two smallest radius ratios, where the printed values are also the smallest and coarsest in the table. Combining the two mixed-edge tables gives 152/156.

6.2 The axisymmetric exact values

SP-160 Table 2.16 collects exact axisymmetric frequencies at $b/a = 0.5$, $\nu = 1/3$. Both mixed-edge entries are MATCH (Table 10): printed 17.51 against 17.598 for inner F / outer C, and printed

Table 9: Mixed edges, inner clamped / outer free, $\nu = 0.30$. Printed values are SP-160 Table 2.30 [5, 6].
[†] the $n = 0$ column at $b/a = 0.9$ is excluded as one named source defect, on the source-internal grounds given in the text. § the production search returned its window endpoint at this cell; an endpoint is not a frequency and is not printed (§3). The re-searched (1, 1) root at $b/a = 0.1$ is WEAK at 1.37% ($\nu = 0.30$) and 1.56% ($\nu = 1/3$) and is scored WEAK, as the stored value was. Paper score 72/76.

(n, s)	$b/a = 0.1$		$b/a = 0.3$		$b/a = 0.5$		$b/a = 0.7$		$b/a = 0.9$	
	printed	solver	printed	solver	printed	solver	printed	solver	printed	solver
(1, 0)	3.14	3.4776	6.33	6.5523	13.3	13.290	37.5	37.498	345	345.17
(0, 0)	4.23	4.2373	6.66	6.6604	13.0	13.024	37.0	36.953	51.5 [†]	§
(2, 0)	5.62	5.6202	7.96	7.9565	14.7	14.704	39.3	39.277	347	347.47
(3, 0)	12.4	12.448	13.27	13.276	18.5	18.562	42.6	42.655	352	351.32
(0, 1)	25.3	25.262	42.6	42.614	85.1	85.033	239	239.84	970 [†]	§
(1, 1)	27.3	§	44.6	44.631	86.7	86.706	241	241.28	2189	2190.1
(2, 1)	37.0	36.943	50.9	50.947	91.7	91.738	246	245.59	2194	2193.9
(3, 1)	53.2	53.207	62.1	62.053	100	100.17	253	252.73	2200	2200.3

13.05 against 13.089 for inner C / outer F. These are three- and four-figure printed values and the agreement is well inside the bar.

Table 10: Exact axisymmetric mixed-edge values, SP-160 Table 2.16 [5], $b/a = 0.5$, $\nu = 1/3$, $n = 0$.

inner-outer	printed λ^2	solver λ^2	rel. (%)	ladder
F-C	17.51	17.598	0.503	CLEAN
C-F	13.05	13.089	0.302	CLEAN

6.3 Southwell’s Table 2.31

Southwell’s values for the free-outer, clamped-inner annulus [7], compiled as SP-160 Table 2.31 at $\nu = 0.3$, are the hardest comparison in this paper and are reported as such: 13/18 under the same 1%/3% bar as every other table (Table 11). No second, table-specific metric is substituted for that score.

The split is a sensitivity, not an accuracy. Southwell’s method assumed round values of the Bessel argument and inverted for the radius ratio, so his tabulated b/a is a computed quantity printed to two or three decimals while his λ^2 is exact by construction. Holding the printed λ^2 fixed and root-finding on b/a instead recovers a CLEAN, $s = 0$ root on *all eighteen* rows. Every row then carries a radius-ratio discrepancy of order 10^{-3} — including the rows that match. For the thirteen MATCH and WEAK rows that discrepancy spans 1.6×10^{-4} to 6.9×10^{-3} , a band that overlaps the misses. What separates the two groups is not the size of the discrepancy but the local sensitivity: the predicted forward error

$$\left(\frac{d\lambda^2}{d(b/a)} \right) \frac{|\delta(b/a)|}{\lambda^2} \quad (5)$$

reproduces the observed relative miss of every row to within a factor of two, 18/18, over three decades of predicted error (Supplementary Material §S.6, Fig. S.1 of that section). The five direct misses are the rows where $d\lambda^2/d(b/a)$ is largest.

This is a unification, and on its own it would be only an internal one. What settles the question is an independent third method.

Table 11: Southwell’s values [7] as compiled in SP-160 Table 2.31 [5], inner clamped / outer free, $\nu = 0.3$, lowest root per n . Score 13/18 under the same bar as every other table. † the production search returned its window endpoint at this cell; the value shown is from a wide re-search and is CLEAN with $s = 0$. All three re-searched rows were, and remain, MISS, so the score is unchanged.

n	b/a	printed λ^2	solver λ^2	rel. (%)	gate
0	0.276	6.25	6.2426	0.119	MATCH
0	0.642	25.0	25.745	2.981	WEAK
0	0.84	81.0	132.99 †	64.185	MISS
1	0.06	2.82	2.8148	0.184	MATCH
1	0.397	9.00	9.0207	0.230	MATCH
1	0.603	21.2	21.258	0.275	MATCH
1	0.634	25.0	25.068	0.274	MATCH
1	0.771	64.0	64.762	1.191	WEAK
1	0.827	121.0	114.17	5.642	MISS
2	0.186	6.25	6.2947	0.715	MATCH
2	0.349	9.00	9.0294	0.327	MATCH
2	0.522	16.0	16.011	0.072	MATCH
2	0.769	64.0	65.583	2.473	WEAK
2	0.81	100	96.526	3.474	MISS
3	0.43	16.0	15.785	1.342	WEAK
3	0.59	25.0	24.964	0.146	MATCH
3	0.71	49.0	45.346 †	7.457	MISS
3	0.82	100	111.04 †	11.041	MISS

Finite elements settle the disputed cell. The most extreme disagreement is at inner C / outer F, $b/a = 0.840$, $n = 0$, where the table prints 81.0 and the solver returns a root near 133. A criterion was fixed before the finite-element run: a lowest FE mode within about 2% of 133 vindicates the solver; one within about 2% of 81 means the solver misses a real mode.

At C–F, $b/a = 0.840$, $n = 0$ the lowest flexural FE mode is $\lambda^2 = 132.549$. The solver gives 132.990 (+0.333%). Southwell prints 81.0 (−63.6%). Nothing in the 40-mode FE spectrum lies below 132.549. A refined mesh (radial divisions $16 \rightarrow 24$, circumferential $256 \rightarrow 384$) reproduces that value as 132.548973 against 132.548990, a relative change of 1.3×10^{-7} . That row is a source error, confirmed externally, and not a solver defect, a conditioning artifact, a basis truncation, or an n -labelling mistake.

The deck’s positive control is the F–C, $b/a = 0.5$, $n = 0$ cell of Table 2.22, where the printed value is 17.7, the solver gives 17.715 and the finite elements give 17.7072: 0.041% against the printed value and 0.044% against the solver. The clamped-edge machinery is therefore correct and the remaining decks are readable.

Eighteen mixed-edge points. Table 12 is the full solver-versus-FE comparison at three C–F geometries, $n = 0$ to 5, every point sign-ladder CLEAN with $s = 0$. All eighteen agree to better than 0.8%; the solver sits *uniformly above* the finite elements, and the gap grows monotonically with n at every geometry, from +0.040% to +0.757%. That is the expected Kirchhoff-versus-Mindlin signature — the 4×4 is thin-plate while SHELL281 carries transverse shear, which softens higher- n modes more — and it is the same one-sided offset reported in [11]. Mixed-sign scatter of this magnitude would have been the worrying result; one-signed monotone growth is not. Two of Southwell’s disputed rows are covered by this set: at $n = 3$, $b/a = 0.820$ the printed 100 is −10.5% from the FE value while the solver is +0.451%; and the MATCH control at $n = 2$, $b/a = 0.522$ has

the printed 16.0 within 0.081% of the FE value.

Table 12: Solver against independent SHELL281 [13] at three inner clamped / outer free geometries, $\nu = 0.30$. Every point is sign-ladder CLEAN with $s = 0$. The offset is one-signed and grows monotonically in n : Kirchhoff versus Mindlin.

n	$b/a = 0.522$			$b/a = 0.820$			$b/a = 0.840$		
	FE	solver	rel. (%)	FE	solver	rel. (%)	FE	solver	rel. (%)
0	14.2651	14.2708	+0.040	104.4191	104.6938	+0.263	132.5490	132.9899	+0.333
1	14.5602	14.5707	+0.072	105.0768	105.3775	+0.286	133.2228	133.6945	+0.354
2	15.9871	16.0115	+0.153	107.0848	107.4618	+0.352	135.2700	135.8325	+0.416
3	19.7175	19.7624	+0.228	110.5422	111.0411	+0.451	138.7636	139.4725	+0.511
4	26.5058	26.5756	+0.263	115.5950	116.2545	+0.570	143.8146	144.7185	+0.629
5	36.3693	36.4717	+0.282	122.4130	123.2632	+0.695	150.5566	151.6958	+0.757

A secondary differential test was also run, comparing the residual radius-ratio discrepancies of Southwell’s table with those of Vogel’s on the same footing. It passed a bar fixed in advance, but its own distribution is heavy-tailed and it does not control for radius ratio, which the two tables sample differently; it is reported in §S.6 as corroboration only. The finite-element result, not that test, is what carries the conclusion of this subsection.

7 In-plane free–free ring

The in-plane reduction of §2.3 is compared with Table 2 of Irie, Yamada and Muramoto [10] at $\nu = 0.3$ and $\beta = 0.2, 0.4, 0.6, 0.8$: the first two frequencies at $n = 1, 2, 3, 4$ together with the $n = 0$ radial and $n = 0$ torsional pairs, 48 cells in all. **All 48 are MATCH**, the largest relative miss being 0.222% at $\beta = 0.8$, $n = 2$, first mode (Table 13; the full 48-row list is §S.4). Both $n = 0$ families are recovered, which matters because the radial and torsional branches interleave differently as β grows. A handful of those MATCH cells classify NOT on the sign-flip ladder ($\beta = 0.4$ and 0.6 in Table 13); scoring is from the relative miss alone, as frozen in §3, and no in-plane finite-element check is claimed.

Comparison here is frequency-only. No in-plane eigenvector cross-check is presented and none is promised: the same comparison was attempted for rectangular in-plane modes in [12] and was not informative, for reasons that apply unchanged to this geometry. The in-plane disk is not claimed: Irie’s “circular” column is $\beta = 0.01$, a tiny-hole control, and until the in-plane 2×2 is built and agrees with it, calling that column $R_i = 0$ would be an assumption rather than a result. The out-of-plane disk of §5 is a genuine $R_i = 0$ result, so the omission is specific to the in-plane motion.

Table 13: In-plane free–free ring against Irie, Yamada and Muramoto Table 2 [10], $\nu = 0.3$, with $\lambda = \omega a \sqrt{\rho(1 - \nu^2)/E}$ of (3). Twelve cells per radius ratio; all 48 MATCH. “worst” names the branch carrying the largest relative miss at that β .

β	cells	gate	worst branch	worst rel. (%)	ladder classes
0.2	12	MATCH	$n = 1$	0.050	CLEAN,FLOOR
0.4	12	MATCH	$n = 2$	0.053	CLEAN,FLOOR,NOT
0.6	12	MATCH	$n = 2$	0.031	CLEAN,FLOOR,NOT
0.8	12	MATCH	$n = 2$	0.222	CLEAN,FLOOR

8 Discussion

What travels from the two companion papers is the Frobenius radial content, the conversion hygiene of §2.4, independent finite elements as a third method rather than a fitted target, and the discipline of freezing a screen before the production sweep. What does not travel is the sector residual bar — which rejects five genuine full-ring roots — the sector subdomain instrument, the rectangular persistence screen, and the Kirchhoff corner jump, which has no θ -edge to live on.

The limits of the present tables are worth stating plainly.

- Some SP-160 tables state $\nu = 1/3$ in their captions and some state no Poisson ratio at all. Every sweep that touches an unlabelled table was run at both $\nu = 0.30$ and $\nu = 1/3$, and the two are never mixed in a single comparison.
- Table 2.35 does not print $n = 4$; the finite elements carry that branch at $b/a = 0.5$ as $(4, 0)$ and it appears here only through the fidelity anchors.
- The $n = 0$ column of Table 2.30 at $b/a = 0.9$ is excluded as a source defect, on grounds internal to Vogel’s own tables. The paper score is 72/76.
- The $(1, 0)$ misses of Table 2.30 at $b/a = 0.1$ and 0.3 are open and named.
- Southwell’s Table 2.31 is 13/18 direct. The sensitivity analysis explains the split; the finite elements, not that analysis, are what resolve the extreme cell.
- The disk tiny-hole control at 10^{-3} is a named FAIL and its bar is not loosened.
- Search-window endpoint returns are labelled and excluded, never quoted as frequencies.
- Clamped–clamped rings, the remaining Vogel edge combinations, simply supported edges, and polar-orthotropic free–free rings [9] are reachable from the same row map with little or no new derivation. They are not in this paper’s production tables, and the cost of adding them is validation rather than theory.

9 Conclusions

A validated annular-sector solver reduces, without a new differential equation, to the classical closed ring and solid disk. The 4×4 radial determinant recovers Vogel and Skinner’s free–free and mixed-edge tables and Irie, Yamada and Muramoto’s in-plane free–free table; the 2×2 recovers the free solid disk, and the 4×4 evaluated at $R_i = 0$ demonstrably does not. Independent finite elements confirm the flexural spine at three free–free geometries, on the disk, and at three mixed-edge geometries, with a one-sided Kirchhoff-versus-Mindlin offset below 0.8% throughout — including the cell where a century-old table prints 81 and the plate vibrates at 133. Piezoelectric identification, using the free–free and clamped-free ring as its elastic forward model, is later work.

Data and code availability

The solver is released under an MIT license as a public GitHub repository, https://github.com/McCune1/plate_solver. For this paper that tree includes the ring and disk entry point (`plate_solver/ring_disk.py`), the sign-flip ladder and window-endpoint labelling, the `TestRingDisk` regression suite that keeps the disk on the 2×2 path, and the tabulated solver-versus-literature JSON under `data/paper3/`. Cluster probe scripts and mixed-edge APDL decks remain in the working package

that produced the tables; they are not part of the Paper 1 or Paper 2 tags. Tables here were captured under `SOLVER_VERSION 2026-07-10.s10`, the same solver as the two companion papers; results under a different version should be checked against the clamped-free regression gates first. Printed literature values are transcribed from the sources cited and are held in one shared module, so that every table cell in this paper is either a transcription or a computed result and none is retyped by hand. The Bessel span is Supplementary Material, §S.1; disk regularity and the rank-3 trap are §S.2; the conversion algebra is §S.3; the full match lists at the second Poisson ratio are §S.4; cluster job numbers are §S.5; the Southwell inverse radius-ratio analysis is §S.6.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Acknowledgments

The computation for this work was performed on The Mill HPC cluster at Missouri University of Science and Technology [14]. GPU resources were not used.

CRedit authorship contribution statement

Garrek McCune: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Daniel Stutts:** Conceptualization, Supervision.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used generative AI tools to assist with manuscript and code organization, drafting, editing, and formatting. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of this publication.

References

- [1] J. Seok, H.F. Tiersten, “Free vibrations of annular sector cantilever plates. Part 1: out-of-plane motion,” *J. Sound Vib.* 271 (2004) 757–772. [https://doi.org/10.1016/S0022-460X\(03\)00414-0](https://doi.org/10.1016/S0022-460X(03)00414-0).
- [2] J. Seok, H.F. Tiersten, “Free vibrations of annular sector cantilever plates. Part 2: in-plane motion,” *J. Sound Vib.* 271 (2004) 773–787. [https://doi.org/10.1016/S0022-460X\(03\)00415-2](https://doi.org/10.1016/S0022-460X(03)00415-2).
- [3] J. Seok, H.F. Tiersten, H.A. Scarton, “Free vibrations of rectangular cantilever plates. Part 1: out-of-plane motion,” *J. Sound Vib.* 271 (2004) 131–146. [https://doi.org/10.1016/S0022-460X\(03\)00365-1](https://doi.org/10.1016/S0022-460X(03)00365-1).
- [4] J. Seok, H.F. Tiersten, H.A. Scarton, “Free vibrations of rectangular cantilever plates. Part 2: in-plane motion,” *J. Sound Vib.* 271 (2004) 147–158. [https://doi.org/10.1016/S0022-460X\(03\)00366-3](https://doi.org/10.1016/S0022-460X(03)00366-3).
- [5] A.W. Leissa, *Vibration of Plates*, NASA SP-160, Office of Technology Utilization, National Aeronautics and Space Administration, U.S. Government Printing Office, Washington, D.C., 1969.

- [6] S.M. Vogel, D.W. Skinner, “Natural frequencies of transversely vibrating uniform annular plates,” *J. Appl. Mech.* 32 (1965) 926–931. The values compared here are as compiled in [5], Tables 2.22, 2.30 and 2.35; the original paper was not re-read for this work.
- [7] R.V. Southwell, “On the free transverse vibrations of a uniform circular disc clamped at its centre; and on the effects of rotation,” *Proc. Roy. Soc. London Ser. A* 101 (1922) 133–153. The values compared here are as compiled in [5], Table 2.31; the original paper was not re-read for this work.
- [8] P.N. Raju, “Vibrations of annular plates,” *J. Aeron. Soc. India* 14 (2) (1962) 37–52. Cited via [5], Tables 2.21, 2.29 and 2.34.
- [9] Y. Narita, “Natural frequencies of completely free annular and circular plates having polar orthotropy,” *J. Sound Vib.* 92 (1) (1984) 33–38.
- [10] T. Irie, G. Yamada, Y. Muramoto, “Natural frequencies of in-plane vibration of annular plates” (Letter to the Editor), *J. Sound Vib.* 97 (1) (1984) 171–175.
- [11] G. McCune, D. Stutts, “Exact wave-function solution and spurious-mode discrimination for completely free annular sector plates,” companion paper, submitted (2026).
- [12] G. McCune, D. Stutts, “Exact wave-function solution of completely free rectangular plates: four-corner Kirchhoff jump and mode screening,” companion paper, submitted (2026).
- [13] ANSYS, Inc., *ANSYS Mechanical APDL*, Release 2025 R1, ANSYS, Inc., Canonsburg, PA, USA (2025). Used for finite-element validation only.
- [14] Missouri University of Science and Technology, “The Mill HPC Cluster,” *The Mill* (2024), Article 1. <https://scholarsmine.mst.edu/the-mill/1>, <https://doi.org/10.71674/PH64-N397>.